

Lecture 27: UMP (size- α) tests, MLR property.

Consider the 4 possible combinations of simple and composite null and alternative hypotheses.

The NP lemma helps construction of MP (size- α) and UMP (size- α) tests, sequentially, in the following manner:

I: Consider the MP (α) test for $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$ (either $> \theta_0$ or $< \theta_0$) - as given
This is simple null ag. simple alt., & the NP lemma can be directly applied

II: Consider the MP (α) test for $H_0: \theta \leq \theta_0$ ($\geq \theta_0$) ag. $H_A: \theta > \theta_0$ ($< \theta_0$)
Repeat I with $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$, ~~any~~ $\theta_1 > \theta_0$ ($< \theta_0$)

Check the power $\text{fn } \beta(\theta)$ of the ~~test~~ test.

It is ~~an~~ MP (α) for $H_0: \theta \leq \theta_0$ ($\geq \theta_0$) ag. $H_A: \theta > \theta_0$ ($< \theta_0$) if the power $\text{fn } \beta(\theta)$ is increasing (decreasing) in θ

III: Consider the UMP (α) test for $H_0: \theta = \theta_0$ ag. $H_A: \theta > \theta_0$ ($< \theta_0$)
As in II, repeat I with $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$, any $\theta_1 > \theta_0$ ($< \theta_0$)

Check the critical region, ω_0 of the test

It is UMP (α) for $H_0: \theta = \theta_0$ ag. $H_A: \theta > \theta_0$ ($< \theta_0$) if ω_0 is independent of the particular choice of $\theta_1 > \theta_0$ ($< \theta_0$).

IV: Consider the UMP (α) test for $H_0: \theta \leq \theta_0$ ($\geq \theta_0$) ag. $H_A: \theta > \theta_0$ ($< \theta_0$)

Repeat I for $H_0: \theta = \theta_0$ ag. $H_A: \theta = \theta_1$, any $\theta_1 > \theta_0$ ($< \theta_0$).

Next, check the critical region ω_0 , whether it is independent of the choice of $\theta_1 > \theta_0$ ($< \theta_0$). If yes, the test

UMP (α) for $H_0: \theta = \theta_0$ ag. $H_A: \theta > \theta_0$ ($< \theta_0$)

Finally, check the power $\text{fn } \beta(\theta)$ of the test, if it is increasing (decreasing) in θ , to have the UMP (α) test for $H_0: \theta \leq \theta_0$ ($\geq \theta_0$) ag. $H_A: \theta > \theta_0$ ($< \theta_0$)

Ex The final type is, composite null agt composite alternative
 $X_1, \dots, X_n$ are iid $N(\mu, \sigma^2)$, σ^2 known, observations. To test

$$H_0: \mu \leq \mu_0 \quad \text{ag.} \quad H_A: \mu > \mu_0.$$

Construct the MP(α) test for $H_0: \mu = \mu_0$ ag. $H_A: \mu = \mu_1 (> \mu_0)$, by the NP lemma.

(this is already done in class)
 and uploaded in an earlier lecture note.

The critical region is $w_0: \sum X_i \geq k$, where 'k' obtained from the size- α condn
 is $k = n\mu_0 + \sigma\sqrt{n} z_\alpha$. Hence the test is given by

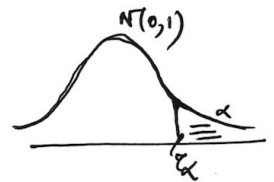
test fn.
$$\phi(x) = \begin{cases} 1 & \text{if } \sum X_i \geq n\mu_0 + \sigma\sqrt{n} z_\alpha \\ 0 & \text{on} \end{cases}$$

Clearly, this test does not depend on μ_1 .

Hence the test is UMP(α) for $H_0: \mu = \mu_0$ ag. $H_A: \mu > \mu_0$.

Now, check the power fn. of the test ϕ , viz.,

$$\begin{aligned} \beta_\phi(\mu) &= E_\mu \phi(x) = P_\mu \left[\sum X_i \geq n\mu_0 + \sigma\sqrt{n} z_\alpha \right] \\ &= 1 - \Phi \left(z_\alpha - \frac{\sqrt{n}(\mu - \mu_0)}{\sigma} \right) \end{aligned}$$



Clearly, $\beta_\phi(\mu) \uparrow \mu$, hence the test ϕ is

UMP(α) for $H_0: \mu \leq \mu_0$ ag. $H_A: \mu > \mu_0$.

If, $H_0: \mu > \mu_0$ ag. $H_A: \mu < \mu_0$, we should first consider the

MP(α) test for $H_0: \mu = \mu_0$ ag. $H_A: \mu = \mu_1 (< \mu_0)$,

check if w_0 (or, ϕ) does not depend on μ_1 and

check whether the power fn. $\beta_\phi(\mu)$ is decreasing in μ .

Monotone Likelihood Ratio (MLR) to obtain a short-cut to the construction of
 UMP(α) tests for certain probability distns.

A family of distns $P = \{f_\theta: \theta \in \Theta\}$ is said to have MLR property in $T(x)$ if,
 $\forall \theta_1 < \theta_2$, $\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} (= L_{\theta_1, \theta_2}(x), \text{ say})$ is non-decreasing in $T(x)$ on the

set of existence of $L_{\theta_1, \theta_2}(x)$.

Note, $L_{\theta_1, \theta_2}(x)$ exists when at least one of $f_{\theta_2}(x)$ and $f_{\theta_1}(x)$ is non-zero.

Ex 1 the one-parameter exponential family has MLR in $T(x)$

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Recall, $f_\theta(x) = h(x) \exp(\eta(\theta)T(x) - \beta(\theta))$, rewritten as

$f_\theta(x) = h(x)c(\theta) \exp(\xi(\theta)T(x))$, where $h(x) > 0$, $c(\theta) > 0$ and $\xi(\theta)$ is strictly monotonic in θ (why?).

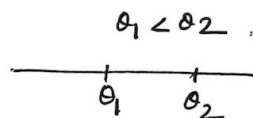
Clearly, the one-parameter satisfies the MLR property if $\xi(\theta)$ is increasing in θ .

What happens if $\xi(\theta)$ is decreasing in θ ?

Ex 2 A non-exponential family of distributions

$X_1, \dots, X_n$ iid from $U(0, \theta)$, $\theta > 0$; then

$$\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} = \left(\frac{\theta_1}{\theta_2}\right)^n \frac{I(x_{(n)}, \theta_2)}{I(x_{(n)}, \theta_1)}$$



$$= \begin{cases} \left(\frac{\theta_1}{\theta_2}\right)^n & \text{if } 0 \leq x_{(n)} \leq \theta_1 \\ \infty & \theta_1 \leq x_{(n)} \leq \theta_2 \end{cases}$$

The distn has MLR in $x_{(n)}$ (when a r.s. of size n is given).

Now, consider ~~comparing~~ testing $H_0: \theta \leq \theta_0$ ag $H_A: \theta > \theta_0$ (UMP size α test).

$\therefore$ The distn has MLR property in some $T(x)$, we can directly write

the UMP(α) test as

$$\phi(x) = \begin{cases} 1 & \text{if } x_{(n)} > c \\ 0 & \text{if } x_{(n)} \leq c \end{cases} \quad \text{why?}$$

(Skipping the UMP(α) test, checking the critical regions & the power f_π).

$$\begin{aligned} 'c' \text{ is } \Rightarrow E_{\theta_0} \phi(x) = \alpha &\Leftrightarrow P_{\theta_0}(x_{(n)} > c) = \alpha \\ \text{ie, } \frac{n}{\theta_0^n} \int_c^{\theta_0} x^{n-1} dx &= \alpha \\ \Rightarrow c &= \theta_0 (1-\alpha)^{1/n} \end{aligned}$$

Next, UMP(α) test directly for the 1-par. exponential family distns.